

Project Report Group 27: Comparing GA, PSO, and ACO for Pathfinding in 2D Mazes

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1 Problem Statement

While standard algorithms like A* excel in static environments and simple grids, they often struggle in unpredictable environments with deceptive dead ends. When an environment suddenly changes, these solvers often have to start over. In contrast, natural computing algorithms use a population to solve problems, allowing the group to adapt together.

The **central theoretical question** of this project is not which algorithm performs best, but how different information architectures shape a population's adaptive behaviour in environments that impose different challenges. We study three paradigms that represent distinct architectures of inter-agent coupling:

- **Genetic Algorithms (GA)** are *decoupled*. Active agents share no runtime information during their evaluation. Knowledge is passed vertically between generations through selection pressure and recombination.
- **Particle Swarm Optimization (PSO)** is *tightly coupled*. Every agent immediately knows the group's best position, producing high social pressure.
- **Ant Colony Optimization (ACO)** is *environmentally coupled*. Agents communicate indirectly through pheromone trails on the grid, so the environment itself becomes the collective memory.

These paradigms give us a natural axis along which to study coupling: no shared memory (GA), continuous global memory (PSO), and distributed environmental memory (ACO). By evaluating them in environments designed to stress different coupling mechanisms, we can characterise when each architecture fails and how effectively a population recovers its diversity. This is the relevance of the work: pathfinding is a controlled, fully observable testbed in which a coupling axis can be isolated, and the conclusions about how shared memory helps or hurts adaptation transfer to any population-based optimiser facing deceptive or non-stationary objectives.

Research Question. To what extent do the information architectures of GA, PSO, and ACO, defined by their varying degrees of agent coupling, shape a population's adaptive behaviour when the environment imposes distinct challenges of deception, non-stationarity, and multimodality?

We decompose this into three subquestions:

- **SQ1 (architecture).** How does the coupling mechanism of each algorithm determine the way its population distributes itself across the search space over time?
- **SQ2 (environment).** How does the type of environmental challenge interact with each coupling architecture to produce distinct failure and recovery patterns?

- **SQ3 (scale).** To what extent does population size modulate adaptive capacity, and does increasing scale compensate for or amplify architectural weaknesses?

2 Hypotheses

We organise our predictions into three families, each tied to a subquestion. We state them explicitly so that the results can confirm or refute them rather than be read selectively after the fact.

H1 (Coupling shapes diversity dynamics; SQ1). Each architecture produces a distinct entropy signature. We expect PSO to show the sharpest entropy collapse on encountering a strong attractor, since the global best pulls all particles toward the same position; GA to maintain the most stable entropy, because decoupled agents cannot lock onto a single misleading path together; and ACO to sit between the two, declining gradually rather than catastrophically because pheromone trails build and erode over time.

H2 (Environmental challenge interacts with coupling; SQ2). Deception (Maze 1) should stress PSO most, because tight coupling accelerates commitment to the misleading attractor. Non-stationarity (Maze 2) should stress ACO most, because pheromone trails on invalidated paths persist until evaporation enables rerouting. Multimodality (Maze 3) should stress PSO most in terms of spatial spread, because social pressure forces the swarm onto a single path even when alternatives are equally valid.

H3 (Population scale modulates but does not reverse architectural effects; SQ3). We expect larger populations to amplify PSO’s weakness rather than compensate for it, GA to benefit most from scale because decoupled random variation improves with additional independent samples, and ACO to be comparatively insensitive to population within the tested range because pheromone dynamics dominate over ant count.

3 Related Work

Sariff et al. [6] applied GA and ACO to robot path planning in static environments, finding that ACO becomes more robust as complexity increases. Shrestha et al. [5] extended this line by benchmarking GA, PSO, and ACO on a common 2D environment with parameters tuned for a waypoint representation. In their setup, a path is formed by connecting a sequence of random coordinate points scattered across an open 2D environment. This sparse representation requires different navigation dynamics than a dense grid. Both compare endpoint performance; neither characterises how the population behaves internally while solving, nor what happens when the environment changes mid-run.

To characterise population-internal dynamics, we track spatial diversity using Shannon entropy. Masisi et al. [4] showed that higher diversity consistently produces better outcomes in optimisation tasks. We adapt this measure to the spatial distribution of agents across the maze, giving a quantitative signal for premature convergence and recovery speed rather than just final performance. Ansari et al. [2] studied efficiency in complex pathfinding, noting that existing heuristic searches such as A* require a large number of iterations to find a path; we use their efficiency framing to structure our benchmarks while investigating the underlying causes of these delays.

Our work differs from Shrestha et al. [5] in four ways. First, we introduce **dynamic environments** by blocking an established path mid-run (Maze 2, Section 4.4) to test adaptation under

disruption, which Shrestha et al. identify as future work. Second, we measure **population-internal dynamics** via position-wise Shannon entropy rather than only endpoint metrics, giving a continuous signal of spatial diversity and premature convergence. Third, we **isolate specific environmental properties** (deception, non-stationarity, multimodality) using purpose-built maze structures, rather than varying complexity monotonically through random obstacle counts. Fourth, we **frame the algorithms as distinct information architectures** and derive failure-mode predictions from that framing, rather than treating them as interchangeable metaheuristics. Together these extensions fill a gap in the comparative literature: how memory architectures degrade and recover when the environment itself changes.

4 Approach

The simulation framework is built from scratch in Python, reusing the core GA and PSO routines from earlier course assignments and adapting them to grid-based navigation.

4.1 The three algorithms at a glance

Before the technical detail, the logic of the comparison is as follows. All three algorithms are population-based optimisers searching the same space (paths from start to goal) under the same budget, and they differ only in how the agents share information. GA agents never communicate within a generation and information survives only through selected offspring; PSO agents all see one global best and are pulled toward it every step; ACO agents leave and read pheromone on the grid, a shared memory that decays over time. Holding the problem, budget, and population fixed lets us attribute behavioural differences to this coupling axis rather than to incidental implementation choices. Each algorithm is detailed below, followed by the shared experimental design.

4.2 Environment and grid model



We model the environment as a 2D grid where agents navigate from a start node to a goal node in the fewest steps. We deliberately chose a discrete grid over a continuous space so that spatial memory (ACO's pheromone) maps directly onto the navigation space, isolating the architectural differences without an interpolation layer that would itself need tuning and would confound the comparison. Walls block movement, and the four-neighbour adjacency means any valid path is a connected sequence of cells, which is the property our GA crossover relies on (below).

4.3 Algorithm parameters and their justification

Because our objective is to strictly compare the different information architectures rather than to maximise absolute performance, we did not tune parameters per maze. Tuning each algorithm to each environment would over-fit and mask the very architectural differences we want to observe; instead we fix parameters from literature sources chosen to match each algorithm's most comparable problem formulation (Table 1). Since no single paper tunes all three on a grid-based maze with cell-by-cell navigation, we combine sources and document each adaptation.

GA. Chromosomes encode a variable-length sequence of cells representing the path from the start node. We initially implemented a binary direction encoding (a fixed-length string of up, down, left, and right moves) so that standard bit-flip mutation could be applied. However, this produced a 0% success rate across all trials. Pilot sweeps on this encoding confirmed the failure was structural rather than a matter of tuning or compute: it persisted across a range of mutation rates and chromosome lengths, and at iteration budgets up to 50,000 generations (these pilot runs are not reported in detail here). The underlying cause is that direction-string crossover

Algorithm	Parameter	Value	Source / rationale
GA	Encoding	Cell sequence (var. length)	Path is the natural representation; preserves validity [3, 7]
	Selection	Roulette wheel	Follows Shrestha et al. [5]
	Crossover	Two-point at common cells (0.5)	Shared-cell splice keeps paths connected
	Mutation	Truncate-and-regrow (0.3)	Rate from [5]; operator adapted to keep paths valid
PSO	Inertia ω	1.0	Directly from Shrestha et al. [5]
	Cognitive c_1	0.1	As above
	Social c_2	0.2	As above ($r_1, r_2 \sim U(0, 1)$ are runtime draws, not parameters)
ACO	Pheromone exp. α	1	Grid-based ACO of Akka and Khaber [1]
	Heuristic exp. β	5	As above
	Deposit Q	2	As above
	Evaporation ρ	0.1	From Shrestha et al. [5]; long pheromone memory
	Initial pheromone	0.8	As above

Table 1: Fixed algorithm parameters and the source or rationale for each. Population size is not listed here because it is a deliberate experimental factor (Section 4.5), tested at 20, 50, and 150 for all three algorithms.

preserves no inheritable invariant, because splicing two valid move sequences almost never yields a valid, continuous path. Therefore, we adopted the cell-sequence representation from Lamini et al. [3] and the variable-length grid chromosome from Tu & Yang [7]. This differs significantly from Shrestha et al. [5], whose fixed-width chromosome encodes random waypoints scattered in an open Euclidean space. Because our environment is a confined grid, the continuous path itself is the natural representation. We retained the roulette wheel selection and the 0.3 mutation rate from Shrestha et al., but adapted the genetic operators to fit our cell-sequence encoding. Two-point crossover is strictly restricted to cells shared between parents to ensure the resulting child path remains connected. For mutation, instead of flipping individual cells, the algorithm cuts the path at a random point and generates a new valid route from that cut to the goal.



PSO. Parameters are adopted directly from Shrestha et al. [5]. Particle positions are projected onto the grid and the velocity update blends inertia, the personal best, and the global best.

ACO. Structural parameters follow Akka and Khaber [1], whose grid-based, cell-by-cell formulation matches ours more closely than the feasible-node graph of Sariff et al. [6] or the waypoints of Shrestha et al. [5]; evaporation and initial pheromone follow Shrestha et al. We abstract colony size away from these baselines because population scale is a tested factor.

Population size. Rather than the disparate per-algorithm sizes reported by Shrestha et al. (70, 40, 20), we align population across all three algorithms and test 20, 50, and 150. Comparing architectures at identical scales isolates coupling effects from a mere population advantage, and treats scale as a deliberate axis (SQ3) rather than a confound.

4.4 Environments

We use three maze types, each engineered to isolate one environmental challenge so that a failure can be attributed to a specific property rather than to generic difficulty. For each type we generate 10 distinct instances that preserve the structural property of that type, and run 10 independent trials per instance. This yields 100 runs per condition and separates algorithmic stochasticity (trial seed) from maze-instance variance (instance seed).

Maze 1 — Shortest Path Trap (deception). A dead end sits on the straight line to the goal. Agents are drawn toward the goal and enter the dead end; escaping requires moving away from the goal, against the distance heuristic. By construction this is a perfect maze: exactly one route reaches the goal, with the trap corridor and a separate dead wedge as dead-end branches. Any agent that reaches the goal therefore does so by the unique (and hence optimal) path. This is the cleanest test of whether an architecture can override a misleading gradient: PSO may stay trapped following a misleading global best, ACO may waste iterations reinforcing pheromone in the dead end, and GA’s random mutation may or may not allow escape.

Maze 2 — Sudden Wall (non-stationarity). The base maze is perfect, with one extra opening added as a shortcut so that two routes exist initially: a short one and a longer alternative. The shortcut is walled at iteration $T = 100$, after the population has settled into it, leaving the long route as the unique path. This isolates recovery from environmental change: the only thing that differs from a static maze is the disruption, so any difference in behaviour after $T = 100$ is attributable to how each architecture handles invalidated memory.

Maze 3 — Parallel Paths (multimodality). Two routes of different length are available from the start (a shorter one and a longer one via a small detour), with no third route. This is the only maze with more than one solution path, and hence the only one on which path optimality can exceed 1. It tests whether a population can spread across multiple optima or whether coupling forces premature commitment to one.

4.5 Experimental design

We run one consolidated three-factor experiment: algorithm $\times$ maze type $\times$ population size rather than three separate studies, so that main effects and interactions come from one coherent dataset. We made this choice because the interesting questions (does the effect of scale depend on the environment? does coupling scale differently?) are interaction effects, which a set of one-factor-at-a-time studies cannot recover.

- **Algorithms:** GA, PSO, ACO (addresses SQ1).
- **Maze types:** Shortest Path Trap, Sudden Wall, Parallel Paths (addresses SQ2).
- **Population sizes:** 20, 50, 150 (addresses SQ3); the range spans Shrestha’s reported 20–100 and extends to 150.

Each of the 27 experiments is evaluated on 10 maze instances with 10 trials each, giving 100 runs per cell and 2,700 runs total. We control random seeds so that all three algorithms are tested on the exact same generated maze instances. The iteration limit is set to $10\times$ the A* shortest-path length for that instance, so the budget scales with intrinsic difficulty and is identical across algorithms on a given instance.

4.6 Evaluation metrics and why they are appropriate

Our research question is about *adaptive behaviour*, not just whether the goal is reached, so we pair internal-dynamics metrics (entropy) with grounding performance metrics (success, optimality). Each metric is chosen to answer a specific subquestion, and we state the link explicitly.

Spatial diversity (Shannon entropy), addresses SQ1. We operationalise the exploration–exploitation balance as spatial diversity: how widely agents are spread across structurally distinct regions at a given iteration. We quantify it with position-wise Shannon entropy

$$H_i = - \sum_{j \in \Sigma} f_{ij} \log f_{ij},$$

where f_{ij} is the fraction of agents in cell j at iteration i . Entropy is appropriate here because it is exactly the quantity that distinguishes the three coupling regimes: a tightly coupled swarm concentrates mass on few cells (low H), while an exploratory population spreads it out (high H). The shape of the curve is our main diagnostic. A sharp cliff indicates rapid collapse, a slow drift indicates lagging environmental memory, a noisy plateau indicates stochastic wandering. For statistical comparison we extract three scalars: *time to 50% entropy loss* (speed of collapse, measured from the peak so we capture collapse only after maximum spread), *diversity floor* (the minimum entropy after the peak, i.e. severity of the worst convergence), and *mean entropy* (sustained diversity over the run).

Adaptation time in Maze 2, addresses SQ2. Iterations from the disruption at $T = 100$ until mean entropy first returns to 80% of its pre-disruption value. A threshold is needed because entropy recovers gradually and “recovered” has no natural cutoff; we apply the same 80% threshold to every algorithm, size, and instance so the comparison is fair. Runs that never recross 80% before the limit are excluded from the adaptation average and captured instead by the success rate, so a slow recoverer is not silently scored as fast.

Success rate (100 runs/cell). Fraction of runs reaching the goal within the iteration limit. High diversity is worthless if the goal is never found, so success rate grounds the entropy analysis in actual task completion and flags cases where an architecture’s coupling prevents finishing.

Path optimality relative to A*. Final path length divided by the A* optimum, computed for successful runs only (1.00 = optimal, A* providing a deterministic baseline). This metric is only informative on Parallel Paths, the one maze with more than one solution route: there it distinguishes algorithms that settle for the slightly longer detour from those that find the shorter route. On the Shortest Path Trap and the Sudden Wall the solution path is unique (perfect-maze construction), so every successful run is optimal by construction and Opt = 1.00 there adds nothing beyond the success rate. We report it throughout for completeness but draw conclusions from it only on Parallel Paths.

4.7 Statistical analysis plan

Because every experiment is repeated 100 times, we report dispersion, not just point estimates, and test the comparisons the subquestions actually ask about. The procedure below was fixed in advance to avoid post-hoc selection and then applied to the full confirmatory run; the complete per-cell confidence intervals and the full table of pairwise tests are provided as supplementary material with the code.

- **Uncertainty.** For each run we report the mean and a 95% confidence interval (bootstrap, 10,000 resamples) for success rate, iterations, optimality, and the entropy scalars. Entropy curves in the figures are shown with shaded 95% bands across the 100 runs.

- **Pairwise differences.** For the headline contrasts (e.g. GA vs. PSO vs. ACO within a maze×size cell) we use the Mann–Whitney U test on the per-run metric, since success and optimality are bounded and non-normal, with Holm correction across the family of comparisons.
- **Factor effects (SQ3).** We fit the metrics with algorithm, maze, and population as factors and test the population main effect and the algorithm×population interaction, which is the formal version of “does scale help, and does that depend on the architecture?”.
- **Effect sizes.** Alongside p -values we report rank-biserial correlation, so that differences are judged by magnitude and not only significance given the large n .

5 Results

All numbers below are from the full confirmatory run of 2,700 trials (100 per cell). We first explain how to read the summary table, then work through the three subquestions in turn, and finally isolate the one result that most needs ruling out as a bug: GA’s complete failure on the deceptive trap.

5.1 How to read the results

Table 2 presents the performance and behavioral data for each experimental cell (algorithm × maze type × population size). To interpret the results systematically, the metrics should be understood as follows:

- **Success rate (SR):** Ranges from 0.00 to 1.00. This indicates whether an algorithm can successfully navigate the maze. A value at or near 0.00 means the algorithm fails to reach the goal within the iteration limit, as seen with GA on the SPT environment; an intermediate value such as GA’s 0.42–0.65 on SW means it solves the maze only some of the time.
- **Optimality (Opt):** Values are ≥ 1.00 , where 1.00 represents the optimal A* shortest path. Higher values indicate less efficient routes. This metric is N/A for cells with an SR of 0.00, because there are no successful runs to evaluate.
- **Entropy ($t_{50\%}$, Floor, and $\bar{H}$):** These quantities are measured in bits. The diversity floor and mean entropy ($\bar{H}$) are bounded below by 0 (indicating all agents have converged on a single cell) and above by the logarithm of total reachable cells. Higher values reflect greater spatial diversity.
- **Adaptation Time (Adapt):** The iterations required to recover to 80% of pre-disruption entropy. This applies exclusively to the Sudden Wall maze and is marked N/A for static environments.

In summary, SR establishes *whether* an algorithm solves a maze, mean iterations and Opt reveal *how efficiently* it solves it, and the entropy metrics illustrate *how the population behaves* throughout the search.

5.2 Diversity dynamics (SQ1)

Figure 1 shows the entropy curves with 95% bands. On the deceptive trap (a) and the Sudden Wall (b) the two coupled architectures behave alike: ACO and PSO both climb to a high diversity plateau of roughly 5 bits, with ACO reaching it faster so its *mean* entropy is marginally the higher of the two (Table 2). GA behaves very differently across these two mazes: on the trap it collapses to a floor near 0.5 bits and never recovers, whereas on the wall it holds a steady,

Algo	Maze	Pop	SR	Iters	Opt	$t_{50\%}$	Floor	$\bar{H}$	Adapt
GA	SPT	20	0.00	2304	N/A	10	0.00	0.39	N/A
GA	SPT	50	0.00	2304	N/A	10	0.07	0.53	N/A
GA	SPT	150	0.00	2304	N/A	10	0.28	0.67	N/A
GA	SW	20	0.42	3070	1.00	67	0.82	2.19	8
GA	SW	50	0.47	2681	1.00	361	1.95	2.90	7
GA	SW	150	0.65	2125	1.00	32	2.41	3.37	7
GA	PP	20	1.00	1	1.01	N/A	3.33	3.33	N/A
GA	PP	50	1.00	1	1.00	N/A	3.87	3.87	N/A
GA	PP	150	1.00	1	1.00	N/A	4.24	4.24	N/A
PSO	SPT	20	1.00	491	1.00	417	3.75	3.31	N/A
PSO	SPT	50	1.00	406	1.00	N/A	5.11	3.85	N/A
PSO	SPT	150	1.00	339	1.00	N/A	6.00	4.25	N/A
PSO	SW	20	1.00	670	1.00	280	3.90	3.18	5
PSO	SW	50	1.00	624	1.00	220	5.07	3.76	4
PSO	SW	150	1.00	586	1.00	230	6.04	4.28	6
PSO	PP	20	0.97	108	1.02	14	1.41	1.06	N/A
PSO	PP	50	1.00	85	1.02	18	1.56	1.10	N/A
PSO	PP	150	1.00	85	1.01	18	1.82	1.23	N/A
ACO	SPT	20	1.00	351	1.00	N/A	4.05	3.56	N/A
ACO	SPT	50	1.00	298	1.00	N/A	5.17	4.24	N/A
ACO	SPT	150	1.00	276	1.00	N/A	6.29	4.83	N/A
ACO	SW	20	1.00	532	1.00	190	3.61	3.74	1
ACO	SW	50	1.00	491	1.00	160	4.76	4.58	1
ACO	SW	150	1.00	464	1.00	N/A	5.72	5.28	1
ACO	PP	20	1.00	79	1.01	N/A	3.38	2.15	N/A
ACO	PP	50	1.00	78	1.00	N/A	4.00	2.43	N/A
ACO	PP	150	1.00	78	1.00	N/A	4.48	2.62	N/A

Table 2: Summary metrics per experimental cell (3 algorithms $\times$ 3 maze types $\times$ 3 population sizes; 100 runs each). **Algo**=algorithm; **Maze**=maze type (SPT=Shortest Path Trap, SW=Sudden Wall, PP=Parallel Paths); **Pop**=population size (number of agents, 20/50/150); **SR**=success rate (0.00–1.00); **Iters**=mean iterations to goal (or to the iteration limit if unsolved); **Opt**=mean path optimality (≥ 1.00 , 1.00=A* optimum); $t_{50\%}$ =iterations from the entropy peak to 50% loss; **Floor**=minimum post-peak entropy (bits); $\bar{H}$ =mean entropy (bits); **Adapt**=iterations to recover 80% entropy after the Sudden-Wall disruption. N/A means the metric does not apply: Opt when SR= 0 (no successful run to measure), Adapt outside Sudden Wall, and $t_{50\%}$ when entropy never falls to half its peak.

moderate level around 3 bits. On Parallel Paths (c) the picture changes again: PSO collapses onto a single route and sits at the lowest diversity (lowest floor in Table 2), ACO spreads across both routes and rises highest, and GA cannot be compared on equal terms because it reaches the goal in its first generation, so its tabulated entropy is merely the initial random spread of one generation rather than a sustained trajectory. The clearest reading of H1 is therefore the opposite of what it predicted: wherever GA actually searches over time (the trap and the wall) it is the least diverse architecture, not the most. These contrasts are statistically unambiguous. On the trap and the wall GA’s diversity floor falls below both coupled architectures with the maximal rank-biserial effect ($r \approx -1.0$, Holm-corrected $p < 10^{-30}$), while on Parallel Paths PSO’s floor sits below ACO’s ($r \approx -0.8$, $p < 10^{-21}$), the signature of the multimodal collapse.

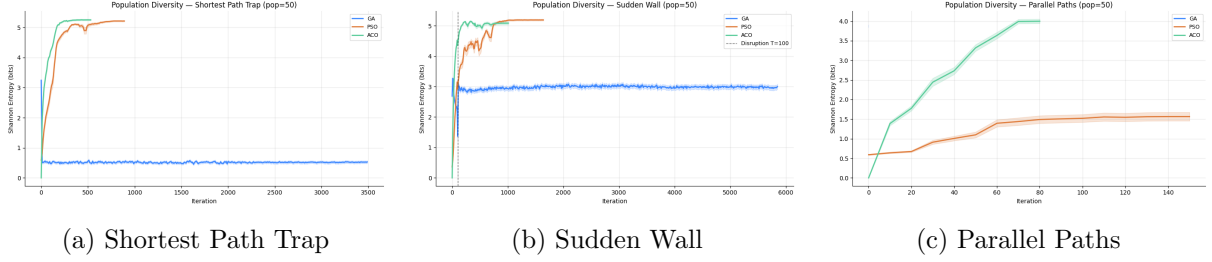


Figure 1: Population diversity (Shannon entropy, bits) over iterations at population 50, with 95% bands across 100 runs. On the deceptive trap (a) and the Sudden Wall (b), ACO (green) and PSO (orange) both climb to a high plateau near 5 bits (ACO faster); GA (blue) collapses to about 0.5 bits on the trap but holds a moderate ~ 3 bits on the wall. The dashed line in (b) marks the disruption at $T = 100$. On Parallel Paths (c), PSO sits lowest having collapsed onto a single route, while ACO rises highest; GA is not visible because it solves the maze in its first generation, leaving a single high-entropy point at the left edge rather than a curve.

5.3 Task performance and environmental challenge (SQ2)

Figure 2 reports success rate by population. GA solves Parallel Paths completely, fails outright on the deceptive trap (SR= 0.00 at every size), and only partially solves the Sudden Wall, where its success climbs from 0.42 to 0.65 as population grows. The coupled architectures solve all three mazes: ACO at 100% everywhere, and PSO at 100% on the trap and wall and 0.97–1.00 on Parallel Paths. The success gap is significant wherever GA struggles: against both coupled architectures it shows the maximal effect on the trap ($r = -1.0$, Holm-corrected $p < 10^{-43}$ at every size) and a smaller but significant gap on the wall that narrows as GA catches up ($r = -0.58$ at population 20 falling to -0.35 at 150; GA success 0.42 [0.32,0.51] rising to 0.65 [0.55,0.74]). On the trap and the wall, optimality is uninformative by construction — each has a single solution path, so any successful run is necessarily optimal (Opt = 1.00) and the performance signal there is carried by the success rate and iteration count. Optimality discriminates only on Parallel Paths, where the table shows PSO most often settling for the slightly longer detour route (Opt up to 1.02) while GA and ACO almost always find the shorter one (Opt ≈ 1.00). One success on Parallel Paths should still be read with care: GA reaches the goal in a single generation (Iters = 1) because that maze has no deceptive structure, so even random initial paths already connect start to goal. We flag this because a raw success rate would otherwise credit GA with an ability it never demonstrates on the harder mazes.

Adaptation after the Sudden-Wall disruption (Figure 3) reverses our hypotheses. We predicted ACO would be slowest because of stale pheromone; instead ACO recovers its diversity in about one iteration at every size, and PSO recovers within four to six, because both maintain a high diversity floor and have therefore already explored the alternative route before the wall appears. GA, the architecture we expected to adapt fastest, is the slowest to recover (seven to eight iterations) and, more tellingly, converts that recovery into a successful solve only 42–65% of the time.

5.4 Effect of population scale (SQ3)

Across all three algorithms and mazes the diversity floor increases monotonically with population (20 $\rightarrow$ 50 $\rightarrow$ 150). Larger populations therefore help every architecture maintain spatial spread; they do not amplify a weakness in PSO as H3 predicted. For PSO and ACO the higher floor translates into fewer iterations to the goal (Table 2). GA benefits most visibly where it has room to: on the Sudden Wall its success climbs from 0.42 to 0.65 as the larger population raises its floor and lets more independent lineages stumble onto the open route. On the deceptive trap,

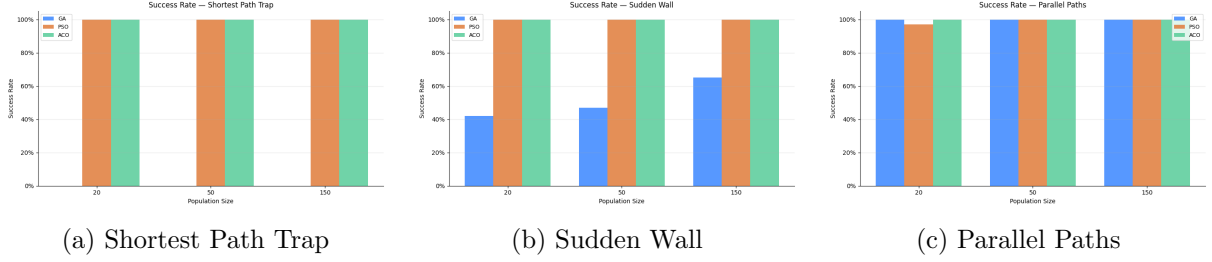


Figure 2: Success rate by population size and maze. GA (blue) solves Parallel Paths, fails the deceptive trap entirely, and partially solves the Sudden Wall with success rising as population grows; the coupled architectures (PSO orange, ACO green) solve all three mazes.

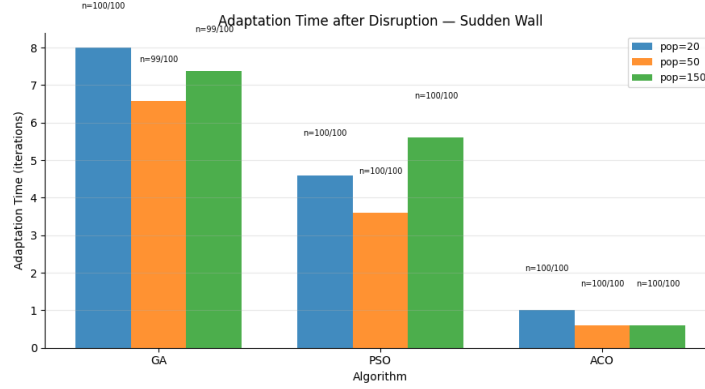


Figure 3: Adaptation time on the Sudden Wall (iterations to recover 80% of pre-disruption entropy). ACO recovers almost immediately and PSO within a few iterations; GA is the slowest to recover its diversity and still reaches the goal only 42–65% of the time.

however, scale is not enough: the floor rises but never crosses the threshold needed to escape, and success stays at 0.00 for every size. A balanced factorial analysis makes this precise (Table 3): the population main effect is significant for every metric, and the algorithm $\times$ population interaction is significant for all but optimality. This is the formal statement of H3’s outcome — scale helps, but by an amount that depends on the architecture, rather than uniformly amplifying PSO’s weakness.

5.5 Did GA fail, or is it a bug?

GA’s outright failure on the deceptive trap is the most surprising result, so we treat the implementation-bug hypothesis as a serious alternative and rule it out using the main experiment itself. First, the same GA implementation succeeds at 100% on Parallel Paths and partially (up to 65%) on the Sudden Wall, returning valid completed paths (Table 2); this shows the encoding, fitness, selection, crossover, and goal test are all wired correctly, because a broken GA would not solve any maze at all. The failure is therefore specific to deception, not a general defect. Second, on the trap GA exhausts its full iteration budget, the same $10\times$ -optimal limit within which ACO and PSO solve the maze, and still reaches 0% at every population size, while its diversity floor rises with population (Table 2) without producing a single success. The bottleneck is thus neither compute nor exploration: GA explores ever more widely yet cannot turn that exploration into an escape, because selection on a distance-to-goal fitness keeps pulling the population back toward the dead end. This inability to override a misleading gradient is a genuine property of the decoupled architecture and is itself the finding rather than a defect to be fixed.

Metric	Population main effect		Algorithm $\times$ population	
	F	p	F	p
Success rate	7.0	9×10^{-4}	5.4	3×10^{-4}
Iterations	9.4	9×10^{-5}	3.0	0.018
Diversity floor	1008.0	$< 10^{-250}$	34.7	3×10^{-28}
Mean entropy	723.0	$< 10^{-250}$	12.1	1×10^{-9}
Optimality (PP)	20.1	3×10^{-9}	1.2	0.33

Table 3: Factor effects from a balanced factorial analysis (Type II ANOVA) over all 2,700 runs. The population main effect is significant for every metric; the algorithm $\times$ population interaction is significant for all but optimality, confirming that larger populations help by an architecture-dependent amount (SQ3). Success rate is binary and iterations are censored at the budget, so for those two we read the F statistic alongside the rank tests rather than on its own. Optimality is restricted to Parallel Paths, the only maze on which it varies.

6 Discussion

6.1 Interpretation: the coupling axis behaves opposite to our prediction

The **headline** finding is largely a reversal of our hypotheses, and it sharpens rather than weakens the contribution. We expected that decoupling would protect diversity (GA most stable, H1) and that tight coupling would be the liability under deception and multimodality (PSO most stressed, H2). The data show almost the opposite where it matters most: GA fails outright on deception and only partially handles non-stationarity, while the coupled architectures sustain diversity and solve every maze. The hypothesis tests **bear this out**. H1 is contradicted wherever GA actually searches over time: on the trap and the wall it is the least diverse architecture, not the most. The one maze where GA records high diversity is Parallel Paths, but there it halts in its first generation, so that value reflects the initial random spread rather than sustained exploration and cannot be read as support for H1. H2 is contradicted on the trap and the wall, where GA rather than PSO or ACO is the worst case, and is supported only for PSO on multimodality, where social pressure does collapse the swarm onto a single route (the lowest diversity floor). H3 is also largely contradicted: scale raises the diversity floor for every architecture and helps PSO rather than amplifying its weakness, although it does help GA most on the one maze where extra exploration can pay off (the Sudden Wall).

The mechanism explains why. GA’s decoupling is not a source of protective diversity but of wasted diversity: its agents explore independently, but with no shared memory the population has no way to retain and build on the discovery that the dead end must be left. Selection on a distance-to-goal fitness actively pulls the population into the trap, and recombination of trapped parents produces trapped children, so the high apparent “diversity” is noise around a bad optimum rather than coverage of useful alternatives. The coupled architectures succeed for the same reason in reverse: PSO’s social term and ACO’s pheromone are shared memory that lets a few lucky discoveries of the escape route propagate to the whole population. ACO is especially robust because its memory lives in the environment and decays, so an invalidated path is forgotten automatically, which is exactly why its Sudden-Wall adaptation is near-instant rather than slow as we feared. GA’s partial success on the wall fits the same account: lacking shared memory it cannot propagate the open route, but a large enough population occasionally contains a lineage that wanders onto it, which is why its wall success rises with scale ($0.42 \rightarrow 0.65$) while its trap success stays at zero. The right axis, then, is not “coupling hurts diversity” but “shared memory converts exploration into exploitable progress, and its absence leaves a population unable to commit even when it has already found the answer.”

6.2 Limitations

We used fixed parameters from the literature and did not tune per maze. This was done deliberate as tuning would over-fit and obscure the architectural differences we set out to measure but it does mean our quantitative results are conditional on these parameter choices. A heavily tuned GA might do better than the untuned one shown here, although it exhausts the full iteration budget on the trap without a single success, which suggests tuning alone would not rescue it there. The dependence is sharpest for ACO’s evaporation rate $\rho = 0.1$, which produces long pheromone memory. We flagged in the proposal that this choice *amplifies* the adaptation effects: a low ρ makes pheromone persist, so the fact that ACO still adapts almost instantly is a conservative result. A higher ρ would only make recovery faster, so our central claim about ACO’s robustness does not depend on a favourable ρ . We say this explicitly because the opposite framing (that we chose ρ to flatter ACO) would be the natural worry, and the evidence runs the other way.

Two further limitations bound the generality of the claims. The parameters adapted from Shrestha et al. were tuned for a different navigation dynamics, not our cell-by-cell grid, so they are a reasonable but not optimal starting point. And no prior work in our reference set evaluates dynamic environments, so the Sudden-Wall results extend the comparative literature into territory where no validated baseline parameters exist; they should be read as a first characterisation rather than a tuned benchmark.

6.3 Future work

Three extensions follow directly. First, an evaporation-rate sweep for ACO and an inertia/social-coefficient sweep for PSO would map how far the robustness we observe depends on memory persistence, turning the qualitative claim into a quantitative boundary. Second, a GA variant with shared memory (e.g. an island model with migration) would test our mechanism directly: if the failure is really the absence of shared memory, adding it should rescue GA on the trap. Third, scaling to larger and procedurally varied mazes, and to more than one disruption per run, would test whether the near-instant ACO/PSO recovery survives repeated non-stationarity or is an artifact of a single, mild disruption.

7 Conclusion

Returning to the research question: the information architecture of a population-based pathfinder strongly shapes its adaptive behaviour, but largely along the opposite of the axis we hypothesised. Decoupling, which we expected to preserve useful diversity, instead leaves GA unable to escape a deceptive dead end at all, even when it exhausts its full search budget, and able to re-route around a new wall only partially and only with a large population. Shared memory, in the form of PSO’s global best and especially ACO’s decaying pheromone, is what lets a population convert exploration into a committed, recoverable solution, with the environmentally coupled architecture (ACO) adapting the fastest to change. Increasing population size helps every architecture by raising the diversity floor, but it does not rescue GA on the deceptive trap. We therefore conclude that this line of work is worth pursuing: a single controlled coupling axis already produces a clear, mechanistic, and partly counter-intuitive story, and the obvious next step — adding shared memory to GA and sweeping the memory parameters of PSO and ACO — is well-posed and likely to be informative.

Author Contributions

Manon Leijser: Implemented the base mazes and the base GA algorithm; contributed to the project proposal and the renewed proposal; incorporated feedback on the renewed proposal; prepared and delivered the flash talk.

Nemo Ingendaa: *[contribution to be added]*

Robert Blaauwendraad: Implemented the PSO and ACO algorithms and the path-encoded GA, redesigned the three maze environments, and built the experiment runner, analysis, and figure scripts; contributed to the writing of the project proposal and final report.

Supplements

Source code, all maze-generation and experiment scripts, the full `runs.csv` (2,700 runs), figures, and a README with installation instructions, package/version pinning (`uv.lock`), a usage manual, and a sample run are available at:

GitHub: <https://github.com/manon-ru/natcomproject>

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